

M Fahrudin Muna

Energy systems optimisation · Stochastic planning under uncertainty · Energy policy assessment

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Nationality: Indonesian | Available from February 2027 (see doctoral timeline below)

PROFILE

Doctoral researcher in Industrial and Systems Engineering at King Fahd University of Petroleum and Minerals (KFUPM), with a first degree in Engineering Physics, the Indonesian professional engineer qualification (Ir.), and over five years of industrial experience in process instrumentation, distributed control systems, and plant design for oil and gas and geothermal projects. Doctoral research covers renewable generation and demand forecasting, multi-stage stochastic generation expansion planning with reliability constraints, and the analysis of research uptake in energy policy. Builds optimisation and forecasting models in Python with Pyomo and Gurobi. Seeks to apply energy-system optimisation and policy assessment to chemical hydrogen carriers and industrial energy systems, and to add process-integrated techno-economic and life-cycle assessment to this toolkit.

EDUCATION

2022 – present **PhD, Industrial and Systems Engineering**
(completion expected January 2027)
King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, Saudi Arabia
Thesis: Energy Forecasting and Balancing: A Multidimensional Approach to Renewable Energy Uncertainty and Load Demand Optimization. Three-paper thesis: paper 1 published (Results in Engineering, 2026); paper 2 manuscript complete; paper 3 in progress (see timeline).

Doctoral completion timeline

Paper 3 (research uptake in energy policy): data collection Sep 2026; bibliometric, content and topic-modelling analysis Oct 2026; results and final draft Nov 2026. Thesis submission early December 2026; defence December 2026; revisions and university clearance by end of January 2027; available to start February 2027.

2021 – 2022 **MSc, Industrial Engineering**
Universitas Gadjah Mada (UGM), Yogyakarta, Indonesia | GPA 3.93/4.00 | LPDP scholarship (Indonesia Endowment Fund for Education)
Thesis: decision-support software for SME supplier selection and inventory decisions, with usability evaluation (published in AIP Conference Proceedings, 2023).

2022 **Professional Engineer Programme, Engineering Physics**
Universitas Gadjah Mada | Insinyur (Ir.), the Indonesian professional engineer qualification

2006 – 2010 **Bachelor of Engineering, Engineering Physics**
Universitas Gadjah Mada | GPA 3.61/4.00
Final project: MATLAB expert-system software for automated weather forecasting.

DOCTORAL RESEARCH

Stochastic generation expansion planning

manuscript prepared for journal submission

- Formulated a multi-stage stochastic mixed-integer linear programme for wind–solar generation expansion across seven generation technologies, with piecewise-linear cost curves, severity-tiered load shedding, and zonal reliability and deliverability constraints.
- Solved a 156-scenario instance on a modified IEEE 300-bus system in Pyomo/Gurobi; the stochastic solution reduces expected unserved energy by 98% relative to the baseline formulation.
- Responsible for conceptualisation, methodology, software, data curation, investigation, visualisation, and drafting.

Renewable generation and demand forecasting

published 2026

- First author of a study of nine renewable generation and electricity consumption datasets combining temporal diagnostics, seasonal decomposition, and LSTM forecasting (Results in Engineering, 2026).
- Quantified the effect of validation-based configuration on LSTM forecast error across the nine datasets.

Research uptake in energy policy

in progress, completion November 2026

- Building Scopus and Overton datasets on forecasting and power-system optimisation research; applying bibliometric analysis, content analysis, and topic modelling to map policy attention and evidence gaps.

METHODS AND TOOLS

Energy systems: generation expansion planning, supply–demand balancing, reliability and adequacy metrics (expected unserved energy, tiered load shedding), scenario analysis, technology portfolio comparison.

Optimisation: multi-stage stochastic MILP, piecewise-linear modelling, scenario generation; Pyomo, Gurobi, IPOPT.

Forecasting and data: time-series diagnostics, seasonal decomposition, LSTM networks, exploratory data analysis; Python, MATLAB, C++.

Policy analysis: bibliometrics (Scopus), policy-document citation analysis (Overton), content analysis, topic modelling, literature synthesis.

Process and plant: instrumentation design, distributed control systems, FEED and detailed-design documentation for gas processing and geothermal facilities.

Project work: nonlinear transportation problem with Pyomo and IPOPT; production control under uncertain seasonal demand for an Indonesian coal producer; CNC parameter selection with moth-flame optimisation and genetic algorithms; simulation-based overall equipment effectiveness assessment.

In development or research interest for application: process-integrated techno-economic analysis and life-cycle assessment; hydrogen-carrier process knowledge; regulatory frameworks for hydrogen, Power-to-X, and CO₂ infrastructure.

RESEARCH INTERESTS FOR IHE-4

Coupling of process, site, and energy-system models for chemical hydrogen carriers (LOHC, ammonia, methanol); effects of electricity procurement and renewable variability on Power-to-X plant cost and utilisation; translation of regulatory requirements into quantitative model constraints; stochastic and multi-objective assessment of carrier and supply portfolios.

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PROFESSIONAL AND RESEARCH EXPERIENCE

- 2022 – 2026 **Introduction Data Science Instructor, Human Factor Lab**
KFUPM university
- Presentation preparation explaining in class.
Conduct some experiment design related to the material, such as anthropometry or regression with real work dataset.
- Apr 2015
– Sep 2018 **System Engineer Level 2, PT Control Systems Arena Para Nusa**
Indonesia | Emerson Automation Solutions representative in Indonesia
- Distributed control system (DCS) engineering for process plants: instrument and I/O lists, DCS design specifications, hardware architecture, control-room layout and network design (TCP/IP and Ethernet), graphic display design, control logic and process diagrams, system safety and SIL requirements.
 - Prepared and executed factory and site acceptance test (FAT and SAT) documentation.
 - Delivered engineering and operator training on DCS operation, maintenance, and engineering.
- Apr – Dec 2014 **Junior Instrument Engineer, PT Ekasapta Energi with PSIT UGM**
Indonesia
- Instrumentation engineering for the detailed design of a 200 MMSCFD gas-compression facility for Pertamina EP Asset 2.
- 2013 – 2014 **Graduate Researcher, Electrical Engineering, KFUPM**
Dhahran, Saudi Arabia
- Research on control-valve stiction compensation using PI-sliding mode control; co-author of the resulting IEEE SSD 2015 paper.
- 2012 – 2013 **Instrument Designer, Centre for Energy Studies (PSE), UGM**
Yogyakarta, Indonesia
- Front-end engineering design of the fluid collection and reinjection system for the 30 MW Karaha Unit 1 geothermal project.
- Sep 2011
– Feb 2012 **Assistant Robotics and Mechatronics Engineer, PT Yaskawa Electric Indonesia**
Indonesia | Robotics and Mechatro Engineering Services Division
- Industrial robot and mechatronic system engineering services.

TEACHING

- 2022 – present **Teaching Assistant and Instructor, KFUPM**
Teaching support in introduction data science course and a human factors lab course. Grading and proctoring for Python, engineering economics, project management, and production and inventory control; instructor for data science
- 2021;
2007 – 2010 **Teaching Assistant, Universitas Gadjah Mada**
C++ programming, digital systems, instrumentation, artificial intelligence, signal and image processing, microcontrollers, and robotics.

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PUBLICATIONS

- Muna, M. F., Malik, A. I., & Syed, M. N. (2026). Dynamic characteristics of renewable energy generation and consumption using LSTM deep learning algorithm. *Results in Engineering*, 30, 109942. <https://doi.org/10.1016/j.rineng.2026.109942>
- Jan, B., Hamid, A., Muna, M. F., & Aljamaan, H. (2025). A survey of sentiment analysis in halal tourism using machine learning and lexicon approaches for extracting sentiment polarity. *Engineering Headway*, 27, 825–844. Trans Tech Publications.
- Muna, M. F., Traspilawati, F., Ushada, M., & Afsana, S. (2023). Usability testing of smart inventory software for EOQ, ROP and SS with recommendation decision in the context of SMEs. *AIP Conference Proceedings*, 2654, 040007.
- Elferik, S., Muna, M. F., Susilo, T. B., & Imran, I. H. (2015). Stiction compensation using PI-sliding mode control for pneumatic valve. *Proceedings of the IEEE 12th International Multi-Conference on Systems, Signals & Devices (SSD)*, 1–6.

Manuscript: Muna, M. F., et al. A multi-stage stochastic framework for wind–solar generation expansion planning with piecewise-linear costs, severity-tiered load shedding, and zonal reliability and deliverability constraints. Manuscript completed 2026; prepared for journal submission.

TRAINING AND CERTIFICATION

Insinyur (Ir.) professional engineer registration, Indonesia (2022) · Gurobi Optimization for Data Scientists (2023) · Applied Data Science Workshop, IEOM KFUPM (2022) · Gas Turbine Operation and Maintenance (2016) · Schneider Electric PLC Programming (2011).

SCHOLARSHIPS AND FUNDING

KFUPM doctoral scholarship; research project funded through the KFUPM Interdisciplinary Research Centers for Intelligent Secure Systems and for Smart Mobility and Logistics (2024 – 2027) · LPDP scholarship, Indonesia Endowment Fund for Education, for the MSc at UGM (2020 – 2022).

LANGUAGES

Indonesian: native. English: fluent; MSc completed and PhD undertaken in English-medium programmes; all publications and teaching in English.

REFERENCES

Available on request.